

ECON 131 - INTERNATIONAL FINANCIAL MARKETS

Pavitra Kanagaraj

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KEY TAKEAWAYS: LECTURES 1 - 3

LECTURE 1 & 2: KEY TAKEAWAYS

- Globalization, MNCs, Forex Market and financial instruments (spot, forward, swap, option)
- **Exchange rate (relative price):**
 - Direct: Domestic currency per unit of foreign currency (\$/£).
 - Indirect: Foreign currency per unit of domestic currency (£/\$).
 - Percentage change formula applies to the **denominator currency**:

$$\% \Delta = \frac{\text{New} - \text{Old}}{\text{Old}}$$

- **Triangular no-arbitrage condition:**

$$\text{Direct Quote}(A/B) = \text{Quote}(A/C) \times \text{Quote}(C/B)$$

LECTURE 2: BID-ASK SPREADS

- **Bid:** Rate at which banks will **buy** the base currency.
- **Ask:** Rate at which banks will **sell** the base currency.
- **Bid-ask spread formula:**

$$\text{Bid-Ask Spread} = 100 \times \frac{\text{Ask} - \text{Bid}}{\text{Midpoint}}$$

- The difference between the ask price and the bid price actually represents two transaction costs. In the first transaction, you buy from a bank at its ask price; then you turn around and sell to another bank at its bid price

LECTURE 3: FORWARD CONTRACTS, PREMIUMS & DISCOUNTS

- **Forward contract:** Agreement to deliver a fixed amount of currency at a predetermined future date at rate F .
- **Market status** (refers strictly to the **denominator currency**):
 - **Forward premium:** $F > S$ (forward price exceeds spot).
 - **Forward discount:** $F < S$ (forward price below spot).
- **Annualized percentage formula since related to interest rates:**

$$\text{Annualized \%} = \frac{F - S}{S} \times \frac{360}{N_{\text{days}}}$$

- **Quick example:** $S = 100/\$, F_{90\text{-day}} = 90/\$$ ($N = 90$, base currency = USD):

$$\frac{90 - 100}{100} \times \frac{360}{90} = -10\% \times 4 = \text{-40.0\% discount (p.a.) on USD}$$

LECTURE 3: KEY TAKEAWAYS

- **Covered Interest Rate Parity (CIP):**

$$1 + i_{\$} = \frac{F_{\$/\pounds}}{S_{\$/\pounds}} \times (1 + i_{\pounds})$$

- The interest rate differential between two currencies approximately equals the percentage spread between the currencies' forward and spot rates. If not, then arbitrage profits exist.

PROBLEM SET 1: HINTS & SAMPLE EXAMPLES

PS1: QUESTION 1

- A US manufacturer exports machinery priced at \$150,000. January 1: Exchange rate is \$1.50/£. June 1: Exchange rate moves to \$1.20/£.
 1. Calculate the percentage change for the currency in the denominator (£).
 2. Did the British Pound appreciate or depreciate against the Dollar? Consequently, did the US Dollar gain or lose value?
 3. Calculate the cost of the machine for a UK buyer in Pounds at both the January and June rates.
 4. Based on your previous answer, explain why US exports become more expensive or cheaper for British customers. How does this affect US trade competitiveness?

HINTS FOR YOUR PS1 PROBLEM 1

- Identify the base currency in \$1.50/£ and \$1.20/£. Which currency sits in the denominator?
- Apply $\frac{\text{New}-\text{Old}}{\text{Old}}$ directly to find the percentage change of the Pound.
- If the Pound changed by a negative percentage, what happened to the purchasing power of the Dollar?
- To find foreign currency price:

$$\text{Price in } \pounds = \frac{\text{Price in USD}}{\text{Exchange Rate } (\$/\pounds)}$$

- Compare the two pound amounts to evaluate whether US goods became cheaper or more expensive for UK consumers.

PS1 PROBLEM 1: PARALLEL EXAMPLE

- **Scenario:** A US airplane component costs \$200,000.

- Jan 1: $S_1 = \$1.25/\text{Euro}$

- Jul 1: $S_2 = \$1.00/\text{Euro}$

- **1. Percentage change of EUR (denominator):**

$$\% \Delta = \frac{1.00 - 1.25}{1.25} = \frac{-0.25}{1.25} = -20\% \quad (\text{Euro depreciated})$$

- **2. Cost to a German buyer in Euros:**

- Jan 1: $\frac{\$200,000}{\$1.25/\text{Euro}} = \text{Euro } 160,000$

- Jul 1: $\frac{\$200,000}{\$1.00/\text{Euro}} = \text{Euro } 200,000$

- Foreign buyers pay **25% more**, dampening US export competitiveness.

PS1: QUESTION 2

- **Problem Prompt:**

- As a foreign exchange trader, you see the following quotes for Canadian dollars (CAD), U.S. dollars (USD), and Mexican pesos (MXN):

- ☐ USD 0.804/CAD

- ☐ MXN 7.432/CAD

- ☐ MXN 8.415/USD

- **Question:**

- Is there an arbitrage opportunity, and if so, how would you exploit it?

HINTS FOR YOUR PS1 PROBLEM 2

- **Check no-arbitrage condition:** Construct the synthetic cross-rate:

$$\text{Synthetic} \left(\frac{\text{MXN}}{\text{CAD}} \right) = \left(\frac{\text{USD}}{\text{CAD}} \right) \times \left(\frac{\text{MXN}}{\text{USD}} \right)$$

- Compare this against the direct market rate of MXN 7.432/CAD.
- If the rates differ, triangular arbitrage exists.
- **Execution strategy:**
 1. Assume a starting capital base (e.g., \$10,000 or CAD 10,000).
 2. Trace the path where you buy the currency in the market where it is cheap and sell it where it is expensive.
 3. Clearly document the cash balances at each of the 3 trade legs and calculate the net profit in your starting currency.

PS1 PROBLEM 2: PARALLEL EXAMPLE

- **Market quotes:** Quote 1: \$1.30/CAD, Quote 2: MXN 150/CAD , Quote 3: MXN 120/\$
- **Step 1: Compute synthetic cross-rate:**

$$\left(\frac{\$}{\text{CAD}} \right) \times \left(\frac{\text{MXN}}{\$} \right) = 1.30 \times 120 = \text{MXN } 156/\text{CAD}$$

- **Step 2: Compare with direct quote:**

Synthetic (MXN 156/CAD) $\neq$ Direct (MXN 150/CAD)

$\Rightarrow$ Arbitrage exists. MXN is underpriced relative to the dollar

PS1: QUESTION 3

- **Problem Prompt:**

- Currently, the spot rate is \$1.50/£ and the three-month forward exchange rate is \$1.52/£.
- The three-month interest rate is 8.0% per annum in the US and 5.8% per annum in the UK.
- Assume that you can borrow \$1,500,000.

- **Question:**

- How would you carry out covered interest arbitrage? Show all the steps and determine the arbitrage profit.

HINTS FOR YOUR PS1 PROBLEM 3

- **De-annualize 3-month interest rates:**

$$i_{3\text{mo}} = i_{\text{annual}} \times \frac{3}{12}$$

- Calculate the covered foreign return factor:

$$R_{\text{covered}} = \frac{F}{S} \times (1 + i_{\text{£},3\text{mo}})$$

- Compare R_{covered} against $(1 + i_{\text{¥},3\text{mo}})$. If unequal, CIP does not hold.
- Structure your solution into 4 distinct steps:
 1. Borrow the lower-yielding currency today ($t = 0$).
 2. Convert to the other currency at spot S .
 3. Invest in the foreign money market for 3 months.
 4. Sell the principal plus interest forward at rate F today.

PS1 PROBLEM 3: PARALLEL EXAMPLE

- **Market data:** $S = \$1.20/£$, $F_{180\text{-day}} = \$1.22/£$.
- Annual rates: $i_{\$} = 6.0\%$ p.a., $i_{£} = 4.0\%$ p.a. Borrow limit: \$1,200,000.
- **De-annualize to 6-month rates:**

$$i_{\$,6\text{mo}} = 6\% \times \frac{180}{360} = 3.0\%, \quad i_{£,6\text{mo}} = 4\% \times \frac{180}{360} = 2.0\%$$

- **Compare gross yields:**
 - Domestic yield: $1 + 0.03 = 1.0300$
 - Covered foreign yield: $\frac{1.22}{1.20} \times (1 + 0.02) = 1.0370$
- Since $1.0370 > 1.0300$, borrow \$, convert to £, invest in £, and sell forward.

PS1 PROBLEM 3: PARALLEL EXAMPLE

1. **Borrow at $t = 0$:** Borrow \$1,200,000 in the US at 3.0% for 6 months.
2. **Convert at spot:** Sell \$1,200,000 at $S = \$1.20/£$ to obtain:

$$\frac{\$1,200,000}{\$1.20/£} = £1,000,000$$

3. **Invest in UK:** Deposit £1,000,000 at 2% for 6 months. Future value:

$$£1,000,000 \times (1 + 0.02) = £1,020,000$$

4. **Forward lock-in:** Sell £1,020,000 forward at $F = \$1.22/£$ today.
5. **Cash settlement at $t = 6$ months:**

- Forward proceeds: $1,020,000 \times \$1.22/ = \$1,244,400$
 - U.S. loan repayment: $\$1,200,000 \times 1.03 = \$1,236,000$
- Arbitrage Profit $\Pi = \$1,244,400 - \$1,236,000 = +\$8,400$

SHORTCUT: QUICK COMPUTATION FORMULA

- You do not need to calculate every leg separately to find the profit.
- General formula:**

$$\Pi = \text{Borrowing} \times \left[\frac{F}{S} (1 + i_{\text{for}}) - (1 + i_{\text{dom}}) \right]$$

- Intuition:** Profit = Principal $\times$ (Covered Return – Borrowing Cost).
- Parallel example check:**

$$\begin{aligned}\text{Spread} &= \frac{1.22}{1.20} (1 + 0.02) - (1 + 0.03) \\ &= 1.0370 - 1.0300 = 0.0070 \quad (0.70\%) \\ \Pi &= \$1,200,000 \times 0.0070 = +\$8,400\end{aligned}$$

SUBMISSION REMINDER FOR STUDENTS

- **Due Date:** In class on **Tuesday**.
- **Format:** **Handwritten work only**.
- **Late policy:** No late submissions or email submissions accepted.
- **Requirement:** Write out all formulas, unit conversions, and operational steps explicitly as shown in these slides.